Approach used to build the home robot demo

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This document provides an overview of the design aspects of the runnable simulations of the home-tidying robot. The robot is intended to navigate around the two-room home and is designed to work with the ROS2 and Gazebo simulation ecosystems.

- The project repo can be found at: https://github.com/deepkarkada/home_robot_demo
- The installation and run instructions can be found at https://github.com/deepkarkada/home_robot_demo/README.md
- The design decisions, challenges and future improvements are highlighted in this document.
- A video recording of the dependency installation, build and launch with the robot navigation can be found here: https://www.loom.com/share/bee7a1c913d44f57a4eb03a138125374

Table 1: Build and run configuration of the test environment

Component	Requirement
OS	Ubuntu 24.04 (Noble)
ROS distribution	ROS 2 Jazzy
Simulator	Gazebo Harmonic
Build system	Colcon

1 Building the robot model

Based on the design specification of the robot, it needs to have a base with a vertical torso with a head and sensors attached. It also needs to have a arm attached on either side of the torso. Based on the spec, the current design is as follows:

- A 4-wheeled differential base (with ros2 differential controller). All 4 wheels make contact with the ground and are actuated.
- The current chassis dimensions are 0.6 x 0.4 x 0.2 m. The torso dimensions are 0.12 x 0.4 x 0.25 m.
- The mass and moment of inertia are modeled accurately based on the dimensions of the objects.

The design is referenced from this tutorial: [1] which has a chassis design and a vehicle model. The four wheels were placed through trial and error (x and y coordinates of the center i.e. pose). The height calculations for the wheel relative to the chassis pose. Friction was not added to the wheel joints since it is not very well supported by the chosen DART [2](default) physics engine (as opposed to friction in ODE which is clearly documented).
The moment of inertia calculations are shown below:

A	B	C	D	E	F	G	H	I	J	K
Rectangle										
Link name	Mass (kg)	Length (m)	Width (m)	Height (m)	Ixx	Iyy	Izz	Ixx = m* (w^2 + h^2)/12		
Chassis	10	0.6	0.4	0.2	0.16666	0.33333	0.43333	Iyy = m* (l^2 + h^2)/12		
Torso	1.2	0.12	0.4	0.25	0.02225	0.00769	0.01744	Izz = m* (l^2 + w^2)/12		
Head on top of torso	0.1	0.02	0.08	0.05	0.00007	0.000024	0.000056			
Lidar	0.1	0.05	0.05	0.05	0.00004	0.000041	0.00004			
Circular disk										
	Mass	Length (m)	Radius (m)	Ixx	Izz	Iyy		Izz (wheel spinning axis)= (MR^2)/2		
Left front wheel	0.5	0.05	0.08	0.000904166666	0.0016	0.000904		Ixx, Iyy (perpendicular axis)= (MR^2)/4 + (ML^2)/12		
Left back wheel	0.5	0.05	0.08	0.000904166666	0.0016	0.000904				
Right front wheel	0.5	0.05	0.08	0.000904166666	0.0016	0.000904				
Right back wheel	0.5	0.05	0.08	0.000904166666	0.0016	0.000904				

Figure 1: Moment of inertia calculations

1.1 Sensors

There are currently two sensors used for the demo: a camera and a LiDAR sensor that publish data to ros2 topics.

1.1.1 Camera

The camera configuration is a front facing camera (per spec) attached at the top of the torso. It has a update rate of 20 FPS. The image data is published to /camera/image topic. It can be enabled during simulations to get front view images that could be used for navigation.

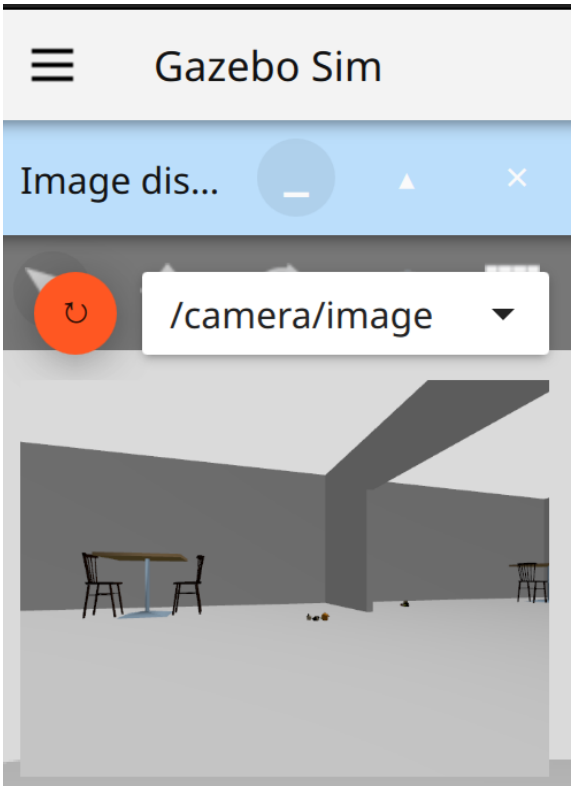

Figure 2: Camera view snapshot

1.1.2 LiDAR sensor

The LiDAR sensor is used for obstacle detection and avoidance. The design is referenced from this tutorial: [3]. It publishes to a topic /scan. It is currently setup as a horizontal scan with 160 degrees field of view.

2 Designing the home layout

The design of the home layout was an interesting challenge in this project. The goal of the design was to model a realistic home with realistic artifacts. The sdf file can be found here: https://github.com/deepkarkada/home_robot_demo/blob/main/ros2_ws/src/home_robot_demo/worlds/home_world_two_rooms.sdf
The current room config is as shown below: More config details are below:

- The walls of the rooms have a length of 6m, width of 6m and height 2.5m.
- Room 1 is centered at x=0, y=0. Hence the north, west, south and east wall boundaries are centered at (0.0, 3.0), (-3.0, 0.0), (0.0, -3.0) and (3.0, 0.0) x,y coordinates respectively with respect to the center.
- The east wall of the first room has an opening that leads into room 2.

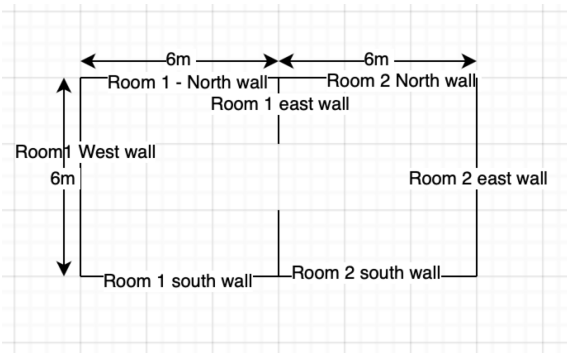

Figure 3: room config (top view)

- There is an empty box modeled in room 1 (spec requirement).
- There is adequate furniture incorporated in the rooms (including a couch, tables and chairs).
- There are small items scattered around the rooms and in the navigation path of the robot such as small toys and blocks for realistic modeling of a home.
  Please note: The models of the toys and block are loaded from Gazebo fuel, which might cause delays in loading of the Gazebo UI and errors such as missing textures. This is temporary and did load correctly in my testing. You might also see an error window pop up asking to quit or wait for GUI to load. Just click on wait and it does load correctly.

3 Navigation strategy

The current navigation strategy is based on LiDAR sensing. It mainly uses the wall avoidance strategy provided in this tutorial [3]. The odometry (current x, y coordinates, distance covered) is published as info logs and also compiled in /tmp/odom_log.csv

4 Tradeoffs made

There were several tradeoffs that had to be made during design:
• Determining the right size (design in general with respect to the room)
• Designing around gazebo constraints - for ex. 4 wheeled diff drive -had to use a 4 wheel diff drive controller from ros 2.

Debugging story: Adding the arms shifted the weight distribution of the body of the robot and caused the motion of the robot to become quite jerky when turning. The

debugging took some effort and the following changes were made:

- The mass of the chassis had to be increased to create smooth motion and reduce risk of tip over.
- It took some effort to see if lack of adequate friction was causing issues. The current gazebo harmonic docs don't provide good documentation on setting friction for the DART engine (docs are available for ODE). However, it turned out that this wasn't an issue.
- The mass of the links in the arms had to be reduced and the damping had to be amped up quite a bit to create a smooth motion.

5 What I would improve with more time

I would definitely attempt the pick and place task if I had more time. Currently I didn't attempt that due to lack of time. I would also use ROS 2 NAV2 or SLAM for navigation.

6 My Build vs. Drift's Build

I've evaluated Drift for home world SDF generation. One issue I saw is that it used an already created home world setup as starting point:

Figure 4: Drift CLI

However, the world looked more realistic with realistic colors and textures. It did seem to have a minor issue with yaw (for ex. for some furniture placement such as dining chairs and bookshelf, shown below):

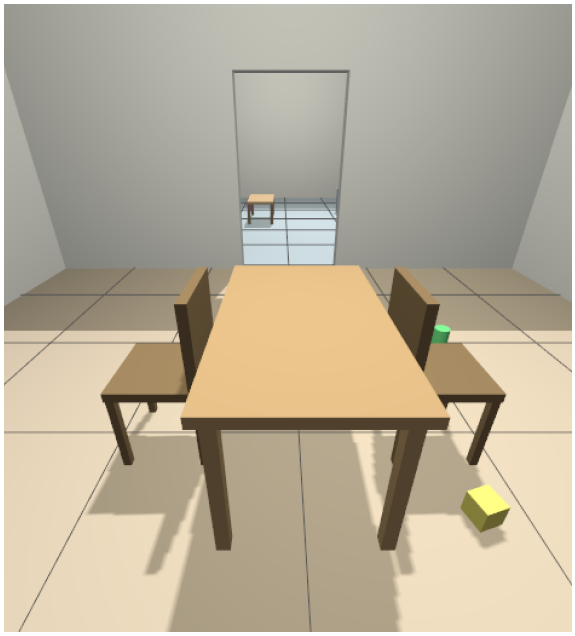

Figure 5: Incorrect yaw calculations

When prompted to create a completely new design (Mobile robot with a mounted SO101 arm), it failed to create an accurate design:

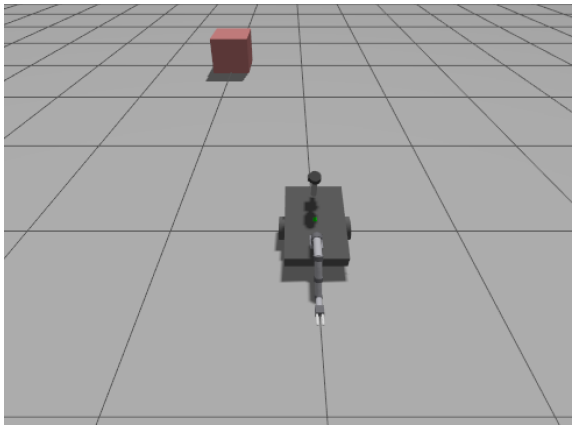

Figure 6: Mobile robot with a mounted SO101 arm

7 References

1 : https://gazebosim.org/docs/harmonic/building_robot

2 : https://gazebosim.org/api/sim/7/physics.html

3 : https://gazebosim.org/docs/harmonic/sensors/